UNIT 5 · 8 hours

Lead Compensator



Compensators, Controllers and State-Space Analysis

Control Systems (22EEG53)
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Learning objectives

- Write the transfer function and frequency response of a lead compensator
- Explain its effect on phase margin, bandwidth and transient response
- Design a lead compensator using the Bode plot
- Implement it with an RC network

Syllabus context

Compensators and controllers: introduction to compensators, lag, lead and lag-lead compensators, P, PI and PID controllers.

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Lead compensator

$$G_c(s) = (1 + aTs)/(1 + Ts),\ a > 1$$

zero at −1/(aT), pole at −1/T

$$\varphi_m = \sin^{-1}[(a - 1)/(a + 1)]$$

maximum phase lead

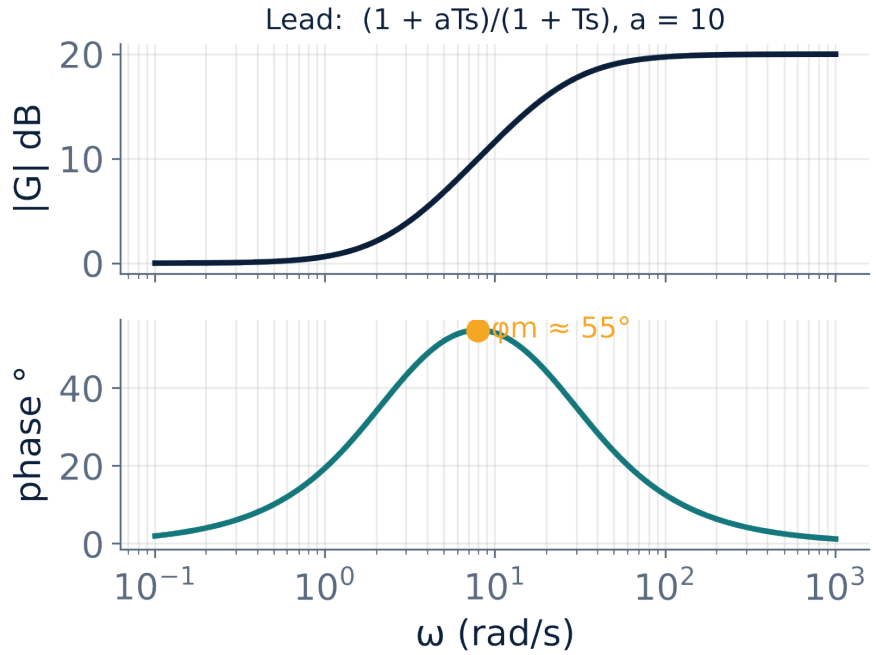
$$\omega_m = 1/(T\sqrt{a})$$

frequency of φm; gain boost 10 log a dB

- Zero closer to the origin than the pole: adds phase lead (positive phase).
- a = (1 + sin φm)/(1 − sin φm).

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Bode plot of a lead network



- Gain rises by 20 log a dB between 1/(aT) and 1/T.
- Phase lead peaks at ωm: 55° for a = 10.
- Place ωm at the new gain crossover frequency.

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Effect of lead compensation

Transient response

Increases PM, damping and bandwidth: smaller overshoot and faster rise time.

Steady state

Little effect (unless gain is raised to recover attenuation).

Cautions

Increases high-frequency gain: amplifies noise; needs extra gain a if implemented with a passive network.

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Bode design procedure (lead)

1

Set K

Choose K for the required steady-state error.

2

Find PM

Draw Bode of KG; read PM.

3

Required lead

φm = PM(required) − PM(existing) + 5°–12°.

4

Compute a

$a = (1 + \sin \varphi_m)/(1 - \sin \varphi_m)$.

5

Locate ωm

Where $|KG| = -10 \log a$ dB; $T = 1/(\omega_m \sqrt{a})$. Check.

$G_c(s) = (1 + aTs)/(1 + Ts)$ with an additional gain a if built from the passive network.

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Numerical: lead design

Problem G = K/[s(s + 2)], Kv = 20 and PM ≥ 50°. Design a lead compensator.

- K = 40: ωgc = 6.17 rad/s, PM = 18°
- Need φm = 50 − 18 + 5 ≈ 37° ⇒ a = (1 + sin37°)/(1 − sin37°) = 4
- Boost 10 log 4 = 6 dB: place ωm where |G| = −6 dB ⇒ ωm = 8.83 rad/s
- T = 1/(8.83 × 2) = 0.0566 s; aT = 0.226 s; Gc = (1 + 0.226s)/(1 + 0.0566s)

Answer: New ωgc = 8.83 rad/s, PM ≈ 50°, Kv = 20 retained

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RC lead network

- $G_c(s) = (1/a)(1 + aTs)/(1 + Ts)$ with $a = (R1 + R2)/R2$, $T = R1R2C/(R1 + R2)$.
- Attenuation $1/a$ is compensated by an amplifier of gain a.
- Example: $R1 = 90\ k\Omega$, $R2 = 10\ k\Omega \Rightarrow a = 10$; $C = 0.1\ \mu F \Rightarrow T = 0.9$ ms; $\varphi_m = 54.9°$ at $\omega_m = 351$ rad/s.
- Matches Experiment 10 of the laboratory.

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Key takeaways

- Lead: $G_c = (1 + aTs)/(1 + Ts)$, $a > 1$; $\varphi_m = \sin^{-1}[(a - 1)/(a + 1)]$
- Increases phase margin and bandwidth: faster, less overshoot
- $\omega_m = 1/(T\sqrt{a})$, placed at the new crossover
- Amplifies high-frequency noise

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Practice questions

Q1. Derive φm and ωm for the lead network.

Q2. G = 20/[s(s + 4)]: design a lead compensator for PM = 45°.

Q3. Compare lead and lag compensators.

Q4. Find T and a for φm = 45° at 10 rad/s.